

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

April/May 2012

EE 303 Microcontrollers & Their Applications

Date: 02.05.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) (i) Convert Decimal fractions to Binary : $(123.78)_{10}$ [03]

(ii) Convert Decimal to Hexadecimal : $(1724)_{10}$

(b) Draw Pin diagram of 8051. [04]

Q - 2 (a) Draw Architecture block diagram of 8051 and Explain each block. [07]

(b) Explain Different Timer modes of 8051 Timer. [07]

OR

Q - 2 (a) Explain ADC 0804 in detail with $\overline{RD}$ and $\overline{WR}$ Timing Diagram. [07]

(b) For Figure 1, [07]

(i) Find the I/O port addresses assigned to ports A, B, C and control register.

(ii) Find the control byte for PA=in, PB=out, PC=out.

(iii) Write a program to get data from PA and send it to both ports B and C.

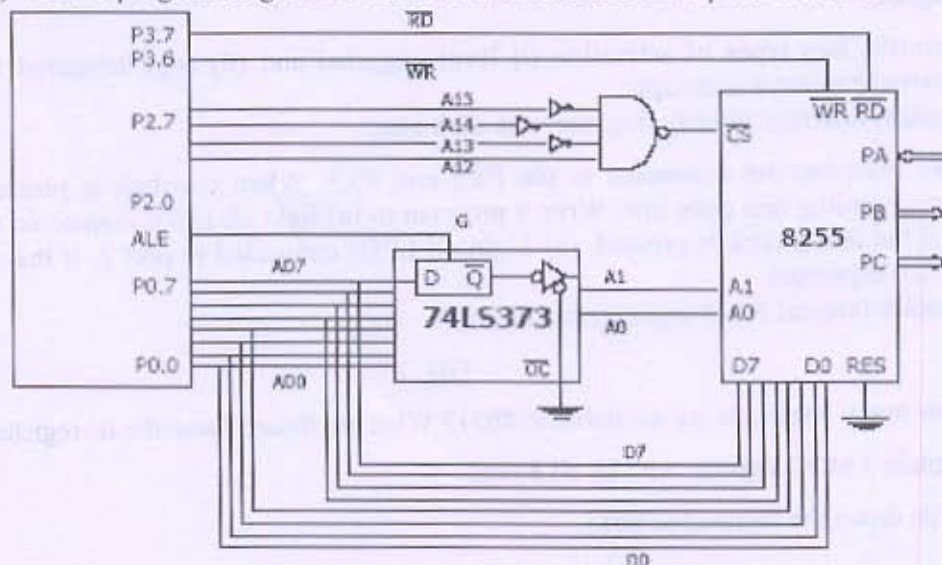
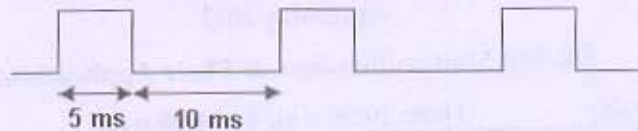


Figure 1

- Q - 3 (a) Generate a square wave with an ON time of 5ms and an OFF time of 10ms on all pins of port 0. Assume an XTAL of 22 MHz. (Use Timer 0 in Mode 1) [10]



- (b) Explain the Programming modes of 8255. [4]

OR

- Q - 3 (a) With a frequency of 22 MHz, generate a frequency of 100 KHz on pin 2.3. Use Timer 1 in mode 1. [06]
- (b) Explain PSW register with its all 8 bits. [04]
- (c) List the three types of buses found in computer systems and state briefly the purpose of each type of bus. [04]

SECTION - II

- Q - 4 Explain Stack and Stack Pointer with POP and PUSH instructions. [07]
- Q - 5 (a) Explain I/O ports of 8051 with its pin circuits. [07]
- (b) Write a program that displays a value of 'Y' at port 0 and 'N' at port 2 and also generate a square wave of 10 KHz with timer 0 in mode 2 at port pin P1.2. XTAL = 22 MHz. [07]

OR

- Q - 5 (a) Explain TCON register with its all 8 bits. [05]
- (b) Explain the Difference between Microprocessor and Microcontroller with necessary diagram. [06]
- (c) Describe two types of activation (i) level triggered and (ii) edge triggered for the external hardware interrupts. [03]
- Q - 6 (a) Explain Interrupt priority Register with all 8 bits. [04]
- (b) Two Switches are connected to pin P3.2 and P3.3. When a switch is pressed, the corresponding line goes low. Write a program to (a) light all LEDs connected to port 0, if the first switch is pressed. (b) Light all LEDs connected to port 2, if the second switch is pressed. [05]
- (c) Explain Internal RAM organization of 8051. [05]

OR

- Q - 6 (a) How many interrupts are available in 8051? What are those? Describe IE register. [05]
- (b) Explain TMOD register with its all 8 bits. [06]
- (c) Write down the features of 8051. [03]
